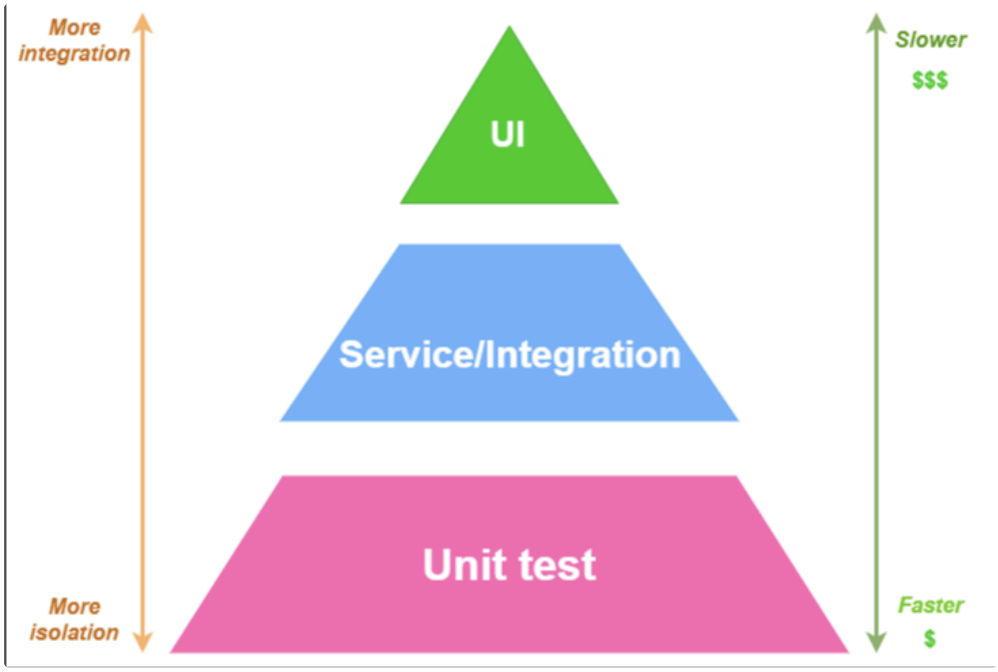
Test Pyramid



Key takeaways:

- The test pyramid is a means of starting the conversation about
  - the distribution of test types that a team should focus on implementing
  - desired test coverage
- Focus on faster and less costly test, whenever possible
- Add levels to the pyramid as needed:
  - static analysis layer
  - manual/exploratory testing layer
- Revisit the pyramid regularly

Characteristics of the test pyramid layers:

Unit test layer

- fastest-running tests
- exhaustive
- easy to get into CI pipeline
- heavy mocking

- tightly coupled to the implementation (white-box tests)
- test failure troubleshooting is easiest at this layer: faults are easily localized

Service/Integration layer

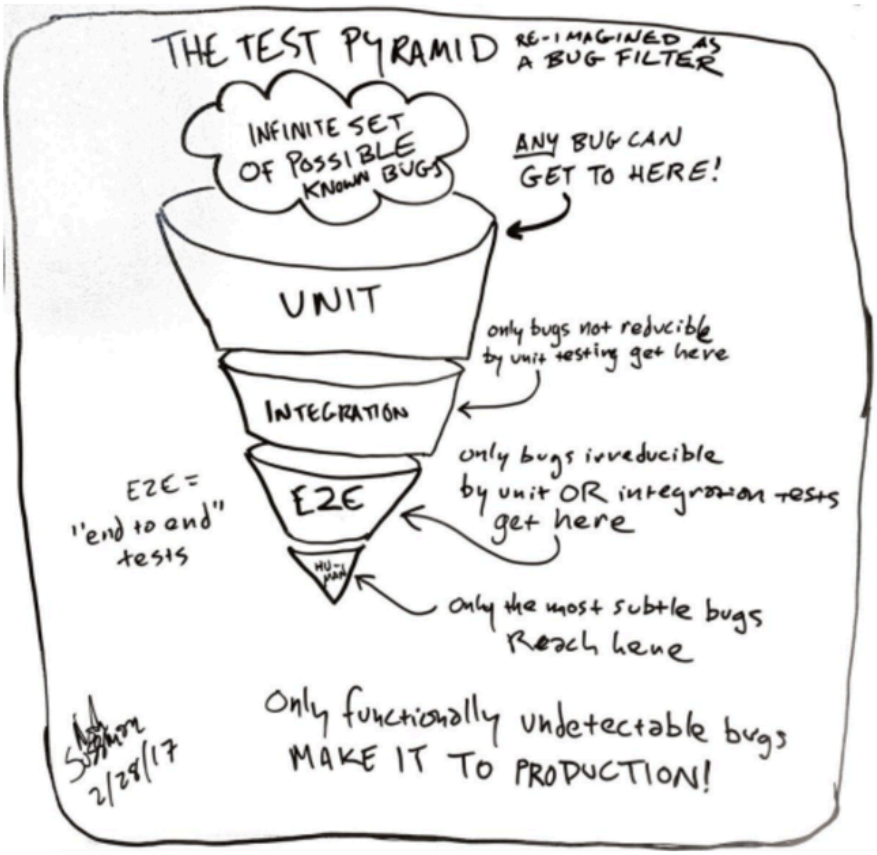
- testing two or more units, possibly even larger subsets of the overall system
- tightly coupled to the implementation (white-box tests)
- still utilize mocking
- test failure troubleshooting is harder at this layer: faults are less easily localized

UI/E2E layer

- tests validate user experience (black-box tests)
- non-exhaustive
- flakiness due to using real infrastructure as opposed to mocks
- test failure troubleshooting is hardest at this layer: faults are even less easily localized

Resources

- Demystifying the test pyramid - Xavier F. Gouchet
  - Noah Sussman, Continuous Improvement
    - Abandoning the Pyramid Of Testing in favor of a Band-Pass Filter model of risk management



- Your automation testing strategy: pyramids, triangles and beyond
  - “I interpret the Agile testing pyramid as a beginning to a discussion about test strategy in an agile team” - Sharon Robson, More Agile Testing:
  - Four basic questions to answer about each level of the pyramid
    - What tests are we writing?
    - How are we using these tests?
    - Why are we using these tests?
    - When and where are we running these tests?
  - have a cloud icon above the pyramid to capture necessary manual and exploratory testing
- the original test pyramid emphasized a relative volume of tests in the various layers (unit > Integration > E2E), which arose from considering the relative costs of writing/running/maintaining tests in each of the layers (E2E > Integration > unit).
- add a level to the Test Pyramid for static analysis
- test environments should be reproducible on local dev workstation, to easily troubleshoot failing CI tests for example
- test coverage graph
- troubleshooting test failures and isolating root causes is easier at the unit test level
- New test types
  - monkey testing
  - fuzz testing
  - property-based testing

- 111. The Test Pyramid Is Dead: A New Era of Software Testing
  - testing vs. quality is captured by the questions “What am I doing? What are we doing?“
  - deciding on the appropriate test type distribution is a team activity